



Event Frequency Layer

Definition Document and Project Plan — MKM-EF-001

MKM Research Labs

Version 2.0 — 22-July-2026

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Status

Document type	Definition Document and Project Plan
Component	Event Frequency Model — MKM-EF-001 (new)
Version	2.0 — supersedes v1.0 of 22 July 2026
Status	Draft for review
Owner	CSO, MKM Research Labs
Proposed tier	1 (Maximum)

This document is tabled for discussion. It proposes a new model for the inventory and a staged plan to remove a pricing defect in the existing flood chain. No code has been written against it.

1 What changed from version 1

Version 1 diagnosed the problem correctly. Version 2 keeps the diagnosis and reworks the plan against what the codebase actually does. Six substantive changes:

#	Change	Driver
C1	The layer is no longer greenfield — it is a port of the existing seismic and fire occurrence pattern onto the storm chain	<code>src/models/seismic/occurrence/</code> and <code>src/models/fire/initiation.py</code> already implement precisely the proposed $\lambda \rightarrow \text{Poisson} \rightarrow \text{event} \rightarrow \text{severity}$ construction. Storm and flood are the outlier, not the pioneer.
C2	Delivery splits into Track A (analytic annualisation) and Track B (ELT $\rightarrow$ YLT); only Track A is in scope	Version 1 went straight to the YLT sampler before any pricing benefit landed. Track A fixes pricing at one node without touching port generation.
C3	Scope is a generic per-peril frequency layer , calibrated for storm first	The wind leg carries the identical defect; fire and seismic already carry their own λ . A flood-only layer would need a second frequency layer within months.
C4	New Section 6: λ provenance is currently circular on synthetic catchments	The “historical” gauge series is generated <i>from</i> the frequency field it would be used to calibrate. This materially changes what Stage 1 can claim.
C5	Version 1’s “critical alignment task” is already broken in a specific, locatable way — sequences are shredded into storms before the denominator is taken	<code>src/models/hazard/io/_load.py:52</code>
C6	Non-functional requirements restated against the actual house rules and audit tooling	Version 1 cited <code>split_audit.py</code> , which no longer exists; specified 100% branch coverage against a house rule of $\geq 99\%$ line; and did not mention the database seam at all.

2 Background and problem statement

2.1 Current state

The simulator generates synthetic storm events and evaluates, at each river gauge, whether a storm produces a flood. Insurance cost is priced as the ratio of flooding storms to total storms. This is a **conditional probability** — $P(\text{flood} \mid \text{storm})$ — and is dimensionless with respect to time.

Two gauges with identical conditional probability therefore receive identical prices even if one experiences eight qualifying storms a year and the other one. The true annualised risk differs by a

factor of eight.

2.2 The defect, with evidence

The defect is not conceptual. It is written out in three places.

Location	Code	Meaning
<code>src/port/src/property/hc/pricing/_process.py:54</code>	<code>spread_bps = (flood_count / num_storms) * 10000</code>	The property PRS spread is $P(\text{flood} \mid \text{storm})$, expressed in basis points
<code>src/port/src/property/hc/pricing/_process.py:250</code>	<code>return_period_yrs = num_storms / flood_count</code>	The assumption stated explicitly: one storm = one year , $\lambda \equiv 1.0$
<code>src/models/hazard/builder.py:109--141</code>	GEV fitted to the peak level of <i>every</i> simulated storm; the resulting exceedance probability assigned to <code>annual_exceedance_prob</code> and <code>annual_hazard_rate_*</code>	A per-event conditional is re-labelled <i>annual</i> at the point of assignment

The same ratio construction appears on the gauge basis leg (`_process.py:119`) and the wind leg (`_process.py:70`, via `_wind_union`).

2.3 What is already correct

The downstream plumbing is rate-shaped and needs no rework:

- `src/models/hazard/gev.py:66` — `compute_term_structure` is already a Poisson compounding, $P(\geq 1 \text{ flood by year } t) = 1 - e^{-\lambda t}$.
- `src/models/hazard/prs_analytical.py:100` — already converts an annual probability to a continuous hazard rate via $-\log(1 - p)$.

Both are being fed a per-event number. **The remedy is a substitution at one node, not a pricer rewrite.**

2.4 The pattern already exists in this repository

Two production models already do what version 1 proposed to invent:

- `src/models/seismic/occurrence/` — `_rates.py` computes a per-asset annual λ ; `_simulate.py:89` draws `rng.poisson(lambda_annual * horizon_years, size=n_sim)` and instantiates events where the count is at least one; severity and damage follow.
- `src/models/fire/initiation.py` — the same construction, labelled “Model A — Poisson initiation”.

MKM-EF-001 should mirror this package shape rather than introduce a parallel abstraction with different conventions.

2.5 Remedy

$$\text{Annual flood frequency}_g = \lambda_g \times P(\text{flood} \mid \text{event}, g) \quad (1)$$

$$\text{Annual exceedance probability}_g = 1 - \exp(-\lambda_g P(\text{flood} \mid \text{event}, g)) \quad (2)$$

$$\text{Pure premium}_g = \lambda_g \times P(\text{flood} \mid \text{event}, g) \times \mathbb{E}[\text{loss} \mid \text{flood}, g] \quad (3)$$

with λ_g the calibrated annual **event** arrival rate — where “event” means the insurance hours-clause event, not the individual storm (Section 5.3).

3 Objectives

#	Objective	Success measure
O1	Price reflects event frequency	AAL differs proportionally between high- and low-frequency gauges with equal conditional flood probability
O2	Honest return periods	<code>return_period_years</code> derived from λ and the conditional, not from the storm count; the 100-year clamp reviewed
O3	Independently validatable components	Frequency backtestable separately from vulnerability; MKM-GH-001 outputs renamed so nothing claims to be annual that is not
O4	Compliance by construction	λ_g provenance recorded and reproducible (BCBS 239), <i>including honest recording of generator-derived provenance</i> ; validation artefacts auto-generated (SR 11-7)
O5	No new technical debt	House rules R1–R6 satisfied; calibration persisted through the <code>database</code> seam, not JSON
O6	Quantified repricing before switchover	Per-gauge and per-property parallel-run delta report produced <i>before</i> the legacy metric is retired

4 Scope

4.1 Locked decisions

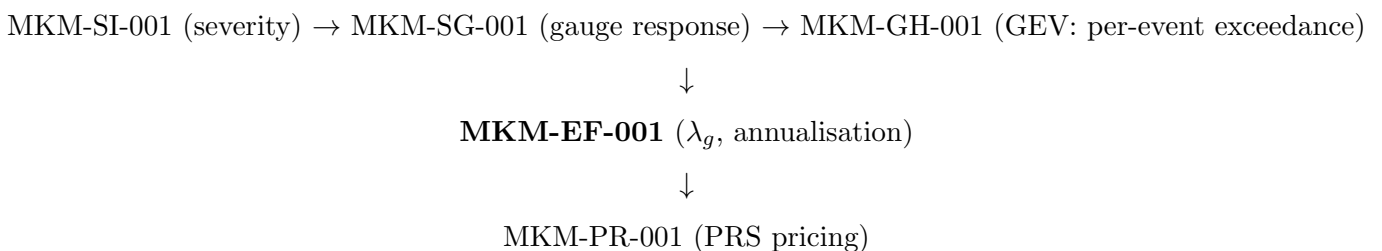
Decision	Choice	Rationale
Delivery	Track A first — analytic annualisation at the hazard-curve node. Track B is a separate approval	Track A is shippable without touching port generation, timeseries generation or the per-property flood-event pipeline

Decision	Choice	Rationale
Peril scope	Generic abstraction with per-peril λ ; calibrate storm first	Wind carries the same defect; fire and seismic already have λ and fold in later
λ source	POT over the gauge record , with a configured regional fallback for short records, the choice recorded in provenance	Matches the target architecture for real gauge data; see Section 6
Chain position	Downstream of storm severity	Nothing upstream changes; one switch point
Model ID	MKM-EF-001 — <i>Event Frequency Model</i>	EF is unused in the model inventory
Tier	1 (proposed)	λ is a linear multiplier on every PRS spread — more material than MKM-SI-001 (tier 2, “not directly into pricing”). Peers at tier 1: MKM-GH-001, MKM-PR-001, MKM-DE-001

4.2 In scope

- Frequency abstraction with Poisson and Negative Binomial implementations, per peril.
- Calibration pipeline: peaks-over-threshold extraction, declustering, arrival-rate estimation, dispersion diagnostics, per-gauge model selection.
- **Event-definition alignment:** the hours-clause event (a storm *sequence*), not the individual storm, becomes the unit of both λ and the conditional.
- The annualisation seam and its wiring into MKM-GH-001.
- Repricing of the property and gauge PRS legs.
- Calibration persistence and provenance through the **database** seam.
- Validation module: dispersion test, count backtest, threshold sensitivity, reconciliation, POT round-trip.
- Configuration schema extension in a new **config/frequency/** package.
- Governance pack: inventory entry, model documentation, registry wiring.

4.3 Chain position



Physically, frequency is a *sibling* of severity rather than a descendant. As an implementation seam, downstream is the right choice: nothing upstream changes, MKM-GH-001 is merely **relabelled** — its outputs become honest per-event conditionals — and `builder.py` is the single switch point.

Inventory `upstream_models` and `downstream_models` must be updated on MKM-GH-001 and MKM-PR-001 accordingly; chain consistency is scanned by full-audit section 4.7.

4.4 Out of scope for this phase

- **Track B:** ELT export, YLT sampler, AEP construction, portfolio aggregation, cross-gauge count correlation.
- Non-stationary or climate-conditioned $\lambda(t)$. Architected for, not implemented.
- Seasonal and non-homogeneous Poisson rates.
- Wind, fire and seismic λ *calibration* — the abstraction accommodates them; only storm is calibrated in this phase.
- Changes to the inundation model, the spatial model or the DEM integration.
- Multi-year contract pricing and discounting.

5 Functional specification

5.1 Package layout

Mirrors `src/models/seismic/occurrence/`. Every file under 300 lines (R2); no callable code in any `__init__.py` (R4); canonical copyright header on every file (R5).

```

1 src/models/frequency/
2   __init__.py           # module docstring + re-exports only
3   datastructures/
4     _rate.py            # FittedRate: lambda, family, params, provenance
5     _diagnostics.py     # dispersion index, log-likelihood, AIC
6   pot/
7     _threshold.py       # threshold selection + stability diagnostics
8     _decluster.py       # minimum inter-event separation
9     _extract.py         # series -> declustered exceedances -> annual counts
10  families/
11    _poisson.py
12    _negbin.py
13    _select.py           # dispersion test + AIC tiebreak; logs any override
14  annualise.py           # p_annual = 1 - exp(-lambda * p_event)  <-- the seam
15  calibrate.py           # pure function of (series, config) -> FittedRate

```

Listing 1: Proposed package layout

`annualise.py` is deliberately tiny and deliberately the only place the composition happens. Frequency logic must not leak into the event simulator or the pricing modules.

5.2 Configuration

A new `config/frequency/` package with `_schema.py` and `_loader.py`, following the hydrated-dataclass pattern of `config/seismic/`. It owns, per house rule R1:

- Target mean exceedance rate band for threshold selection.
- Declustering window (minimum inter-event separation).
- Dispersion-test critical value; AIC tiebreak policy.
- Minimum record length for per-gauge calibration; regional fallback λ .
- Return-period grid for output tables.
- Per-peril family overrides, with the override itself logged (SR 11-7).

It also absorbs an existing orphan: `src/models/intensity/distribution/_core.py:198` carries `storms_per_year: float = 20` as a default argument — an undocumented frequency assumption living inside the severity model, and an R1 breach. That parameter moves here and the severity model takes it as an injected value.

5.3 Event-definition alignment

This is not a hypothetical risk. It is already wrong, and locatably so.

MKM-SS-001 produces **sequences** — one to five related storms fitted inside a 168-hour insurance hours clause (`EVENT_WINDOW_HOURS` in `config/port/_storm.py`). But `src/models/hazard/io/_load.py:52--61` flattens every sequence into its member storms before the hazard model sees them. `num_storms` is therefore one to five times the event count, and today's denominator is **neither events nor years**.

Remedy:

1. Add an event-granular loader alongside the existing flattening one, which stays for consumers that genuinely want individual storms.
2. Evaluate the conditional per sequence — the gauge response over the whole 168-hour window, that is the maximum peak level across member storms, not one row per storm.
3. Quote λ in **hours-clause events per year**, matching that unit exactly.

This step alone reprices, before λ is applied, because the denominator changes. It is sequenced first precisely so its effect can be measured in isolation from the λ effect.

5.4 Frequency abstraction

A single interface, per peril, following the seismic rates/simulate split:

- `fit(counts: AnnualCountSeries) -> FittedRate`
- `annual_rate() -> float`
- `sample_annual_count(rng) -> int` (*unused in Track A; required by Track B*)
- `diagnostics() -> FrequencyDiagnostics`

Implementation	Distribution	Selected when
PoissonFrequency	Poisson(λ)	Dispersion index ≈ 1 (default)
NegativeBinomialFrequency	NegBin(r, p)	Overdispersed counts (variance > mean)

Selection is data-driven per gauge via the dispersion index with an AIC tiebreak, recorded in the calibration output. A configuration override may force a family; the override is logged with its justification.

5.5 Calibration pipeline

Per gauge, from the gauge level record:

1. **Threshold selection** — the threshold yielding a target mean exceedance rate, anchored on bankfull discharge where available. Diagnostics: mean-excess and threshold-stability.
2. **Declustering** — enforce a minimum inter-event separation so peaks are independent. Record the rule applied.
3. **Arrival rate** — $\hat{\lambda} = \text{exceedances} / \text{years of record}$; the annual count series is retained for dispersion testing.
4. **Dispersion test and family selection.**
5. **Severity cross-check (validation only)** — fit a GPD to exceedance magnitudes and compare against the simulator's implied severity at that gauge. Divergence beyond tolerance raises a validation flag; it never silently recalibrates anything.

Prior art to fix, not to write from scratch. `src/models/statistics/timeseries.py:103-122` already computes a per-gauge `frequency_per_year` for alert, warning and severe. It counts **days over threshold**, not declustered events — peaks-over-threshold with step 2 missing — and therefore overstates λ by roughly the mean event duration in days. Stage 1 is a fix-and-promote of this calculation into the new package, with the existing consumers repointed.

Provenance (BCBS 239). Every fitted rate carries: source dataset identifier and version, record period, threshold, declustering rule, family selected and why, fit timestamp, code version, configuration hash, and a **provenance class** — `observed`, `generator-derived` or `regional-fallback`. Calibration is a pure function of (data, configuration): re-running with identical inputs must reproduce identical outputs, regression-tested against a golden master.

5.6 Persistence

Fitted rates and their provenance are written and read **exclusively through the `src/database/seam` (R6)**. No JSON artefact is created at any point — full-audit section 4.5 is walking JSON writes toward zero tolerance, and a new model must be born database-native.

Deliverables: a new table plus Alembic migration following the existing gauge hazard-curve migration, with matching entries in the ORM model and column maps, and accessor functions in the database package API.

5.7 Wiring — three edits

Only the third reprices via λ ; the first reprices via the denominator.

#	File	Change
1	<code>src/models/hazard/io/_load.py:52</code>	Add the event-granular loader; sequences stop being shredded before the denominator is taken

#	File	Change
2	src/models/hazard/builder.py:109--141	GEV fits per-event peaks; outputs renamed to <code>event_exceedance_prob</code> ; <code>annual_hazard_rate</code> populated via the annualisation seam
3	src/port/src/property/hc/pricing/_process.py:54,119,250	Property spread, gauge basis leg and return period become λ -based

5.8 Pricing outputs (Track A)

- **AAL and pure premium** — $\lambda_g \times P(\text{flood} \mid \text{event}, g) \times \mathbb{E}[\text{loss} \mid \text{flood}, g]$.
- **Annual exceedance probability** — $1 - \exp(-\lambda_g p_{\text{event}})$, replacing the current direct assignment.
- **Return periods** — derived from the annualised probability, with the compensating clamps reviewed.
- **Event exceedance frequency** — valid below the one-year return period, which matters at frequent-flood gauges and cannot be expressed by a probability at all.

Occurrence and aggregate exceedance probability curves require the year loss table and are **Track B**. Track A can state OEP analytically under Poisson, $1 - \exp(-\lambda S(x))$, but not empirically, and AEP not at all.

5.9 Extension points

Architected now, built later: λ as a process rather than a constant (the interface takes an optional covariate or time argument); `sample_annual_count` present from day one though unused by Track A, so Track B needs no interface change; a per-peril λ registry so wind, fire and seismic attach without restructuring.

6 The λ provenance problem

Finding. On synthetic catchments, calibrating λ from the gauge “historical” record is circular.

`src/models/statistics/synthetic.py:122--124` generates the daily series by injecting severe exceedances at exactly the rate implied by the gauge’s `FrequencyExceedLevel3` field:

```
1 severe_floods_expected = max(freq_exceed_level3, 1)
2 severe_flood_prob_per_day = severe_floods_expected / (years * 365)
```

and `FrequencyExceedLevel3` is itself produced by `src/port/rand/shared/gauge/gauge_field_generators.py:168--170` as a `random.choices(...)` draw over a five-year window.

A peaks-over-threshold extraction over that series therefore recovers the injected rate. **The only per-gauge frequency number the platform currently holds is a random integer.**

Consequences.

1. Version 1’s stated risk — “short or gappy records give a noisy $\hat{\lambda}$ ” — is the wrong risk. The real one is that λ has no observational content on synthetic catchments.

2. It does not invalidate the POT pipeline. POT is exactly what would be run against real gauge series, and building it now is correct.
3. It does change what Stage 1 may claim. The provenance class must record **generator-derived**, not **observed**, and the validation report must say so.

Approach adopted.

- **Now:** build POT as specified. On synthetic catchments, validate it by **round-trip** — the extracted $\hat{\lambda}$ must recover the injected rate within tolerance once declustering is accounted for. This is a genuinely strong test of the extraction code, and it is labelled as such rather than presented as calibration.
- **Follow-on** (separate approval; requires port regeneration): make `FrequencyExceedLevel3` a *modelled* field rather than a random one, giving the gauge generator a catchment-level λ prior so synthetic frequencies are physically plausible and spatially coherent. Gauges on the same reach should not carry independent λ .

7 Non-functional requirements

Restated against the project coding rules.

Rule	Requirement for this project
R1 — no parameters outside <code>config</code>	All thresholds, declustering windows, selection criteria, fallback λ and return-period grids live in <code>config/frequency/</code> . The orphan <code>storms_per_year=20</code> is absorbed.
R2 — no file over 300 lines	Package layout adopted from the outset.
R3 — $\geq 99\%$ coverage, per stage	Verified at the end of <i>each</i> stage. Version 1 specified 100% branch coverage; aligning to the house rule avoids a bespoke standard for one package. Plus property-based tests for mean and variance recovery, and a golden-master calibration regression.
R4 — no functions in <code>__init__</code> <code>--py</code>	Re-exports only.
R5 — canonical copyright header	On every new source file; enforced by the self-healing header audit.
R6 — data store only via <code>database</code>	Zero direct SQL, ORM or connections; zero JSON artefacts.

Additionally: seeded RNG throughout, with calibration and simulation reproducible from (data version, configuration version, seed); a structured run log carrying inputs, configuration hash, code version, outputs and validation flags; and verification through the full audit — *not* `split_audit.py`, which version 1 cited and which no longer exists. Relevant full-audit sections: 4.3 path definitions, 4.5 JSON files, 4.6 database usage, 4.7 model chain consistency.

8 Project plan

Leaves first; each stage independently shippable and tested; coverage checked at every stage boundary.

Stage	Content	Reprices?
1	Event-definition alignment; POT extraction with declustering; frequency_per_year fix-and-promote; λ persisted through the seam with provenance	Denominator only
2	Poisson and NegBin families; dispersion selection; diagnostics; override logging; POT round-trip validation	No
3	Annualisation wired into builder.py ; legacy metric behind a deprecation flag; parallel-run repricing report	Yes — via λ
4	Property and gauge legs in _process.py ; return periods; clamp review	Yes
5	Governance: inventory entry, model documentation, registry wiring, auto-generated validation report, monitoring job	No
6	<i>Separate approval:</i> Track B — ELT export, YLT sampler, OEP and AEP, portfolio aggregation, wind λ , seasonal rates, cross-gauge dependence	—

8.1 Stage acceptance

- **S1** — POT series reproducibly generated for all gauges; provenance complete; event-granular loader covered; no pricing path yet consumes λ .
- **S2** — families pass property tests for mean and variance recovery; calibration deterministic under fixed inputs; per-gauge selection report generated; round-trip recovers the injected rate within tolerance.
- **S3** — parallel-run report produced, quantifying repricing per gauge; legacy metric still available behind the flag; analytical and empirical AAL reconcile within Monte Carlo error.
- **S4** — property-level repricing quantified; clamp decisions recorded; user interface and end-of-day consumers verified against the renamed fields.
- **S5** — validation report reviewed and signed off; monitoring scheduled; legacy metric formally deprecated.

9 Migration landmines

These will bite between Stage 3 and Stage 5. Each needs a decision before Stage 3 lands.

#	Landmine	Detail	Action
L1	Magnitude of the reprice	$\lambda \equiv 1$ today. A plausible λ of 2–4 events per year scales spreads by roughly that factor at small p . This is a 2–4 $\times$ move on <i>every</i> PRS quote.	The Stage 3 parallel-run report is the gating business artefact. No switchover without it.
L2	Compensating hacks become distortions	builder.py:87 clamps the exceedance probability up to 0.01; MAX_RETURN_PERIOD: 100 caps return periods. Both exist <i>because</i> return periods were fake.	Once return periods are real, both must be reviewed or removed. Removing them widens the tail — quantify alongside L1.
L3	num_storms is persisted and rendered	It is a stored field read by the user interface, the blotter and end-of-day. Changing the denominator to events changes its meaning and value.	Either keep num_storms and add num_events, or rename with a migration. Decide before Stage 1 ships.
L4	End-of-day history spans the cut	Stored annual_hazard_rate* values change semantics <i>and</i> magnitude; historical series become non-comparable across the switchover date.	Mark the cut in the end-of-day record; do not silently backfill.
L5	Per-storm flood events collapse	Under an event definition, several storms in a sequence become one event; per-property flood counts drop even before λ is applied.	Measure separately at Stage 1 so it is not confused with the λ effect at Stage 3.
L6	Wind leg diverges	The wind union also divides by num_storms. If flood is annualised and wind is not, the BOW and BAW union and joint legs become internally inconsistent.	Either annualise both at Stage 4 or explicitly freeze the wind leg on the legacy metric with a recorded justification.

10 Governance and registration

Per the new-model checklist, and noting that a new model document must be wired into **five registries** plus the Makefile filter:

1. Inventory entry MKM-EF-001 — tier 1 (proposed), category Hazard, source module `src/models/frequency/`, upstream MKM-GH-001, downstream MKM-PR-001.
2. Corresponding upstream and downstream amendments on MKM-GH-001 and MKM-PR-001; the chain is hand-maintained and scanned by full-audit section 4.7.
3. `docs/models/event_frequency/event_frequency.tex` and Makefile, with the standard sections and the generated test-results and sensitivity inputs.

4. Registry wiring: five registries plus the Makefile filter.
5. Assumptions and limitations recorded — including, explicitly, the provenance limitation of Section 6 and the stationarity assumption.

Validation plan (SR 11-7). Conceptual soundness; per-gauge count backtesting on a held-out split; dispersion coverage reporting (share of gauges selecting NegBin); threshold sensitivity ($\hat{\lambda}$ and AAL under a one-band threshold perturbation); reconciliation of analytical against empirical AAL; the POT round-trip; annual recalibration with a drift alert; and an effective-challenge artefact emitted on every run.

Note that benchmarking return-period levels against published national flood-frequency estimates — version 1’s validation test 6 — is **not meaningful** while catchments are synthetic. It is retained in the plan but flagged as pending real gauge data.

11 Risks and assumptions

Risk or assumption	Impact	Mitigation
λ has no observational content on synthetic catchments	Pricing is precise but not accurate; validation claims could overstate	Provenance class recorded on every fitted rate; round-trip framed as extraction validation, not calibration; real-data path built and ready
Repricing of 2–4× on every quote	Client and book impact	Stage 3 parallel-run report before switchover; staged rollout; legacy metric retained behind a flag
Event-definition change reprices before λ does	Two effects confounded in the impact analysis	Sequenced deliberately: Stage 1 isolates the denominator effect, Stage 3 the λ effect
Threshold choice materially moves $\hat{\lambda}$	Pricing instability	Sensitivity test; threshold governance in configuration with a change log
Overdispersion ignored where present	Understated tail	NegBin selected wherever the dispersion test rejects Poisson
Stationarity assumed	Understated forward risk under climate change	Explicit limitation in the model documentation; drift monitoring; Track B roadmap
Wind leg left on the legacy metric	Internally inconsistent peril comparison	Decide explicitly at Stage 4; do not leave implicit
Clamp removal widens the tail	Compounds the repricing impact	Quantify jointly with the repricing report, not separately

12 Glossary

AAL

Average Annual Loss; expected loss per year.

AEP

Aggregate Exceedance Probability; the probability that total annual loss exceeds x . *Track B*.

OEP

Occurrence Exceedance Probability; the probability that the largest single event loss in a year exceeds x . *Track B*.

EEF

Event Exceedance Frequency; the annual frequency, not probability, of events exceeding a level.

ELT / YLT

Event Loss Table and Year Loss Table. *Track B*.

Event

Here, the insurance hours-clause event — a storm *sequence* within the configured event window — not an individual storm.

POT

Peaks Over Threshold; extreme-value sampling of all independent exceedances above a threshold.

GPD

Generalised Pareto Distribution; the standard model for peaks-over-threshold exceedance magnitudes.

Dispersion index

Variance divided by mean of annual counts; near one supports Poisson, above one indicates overdispersion.

NHPP

Non-Homogeneous Poisson Process; a time-varying arrival rate. *Track B*.

Provenance class

observed, generator-derived or regional-fallback; records what a fitted λ is actually based on.